# Differential Methylation of Expression-Associated CpG Loci in an Enhancer Region is Associated with Morning Cancer-Related Fatigue Severity

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## Background and Purpose

Cancer-related fatigue (CRF) is the most common symptom associated with cancer and its treatments. CRF has a negative impact on patients' ability to tolerate treatments and on their quality of life (QOL). Given its high occurrence and negative impact, effective treatments need to be developed for this devastating symptom. One of the major knowledge gaps for CRF is a lack of knowledge of its underlying mechanisms. Increased knowledge of CRF mechanisms could identify potential therapeutic targets.

CRF is a complex phenotype, which is the result of environmental (e.g., demographic, clinical, social) and molecular factors. By integrating data from multiple sources, we will have increased power to identify and better interpret the omics-phenotype relationships relative to an analysis that uses only a single source of omics data. We will use multi-omics strategy to obtain a more complete biological model of the molecular factors associated with the severity of morning CRF.

Gene expression is regulated, in part, by epigenetic mechanisms. A major unanswered question in genomics research is the functional contribution of epigenetic variation on gene expression. One method to evaluate for the potential functional effect of a methylation variation is to test for an association between levels of methylation and gene expression from the same samples. These expression-associated quantitative trait methylation (eQTM) loci may contribute to the regulation of gene expression (also called expression associated CpG loci, eCpG). These associations may be local (e.g., methylation located in the promoter region of a gene) or remote (e.g. methylation loci in a distant enhancer regions of a gene or on a different chromosome). There is growing interest in the integration of these data modalities and evaluating for eQTMs.

In the parent study (CA233774), we are performing multi-staged data-integrated multi-omics analysis (Harris et. al, PMID 33827270) to identify potential mechanisms for CRF (e.g., Kober et. al. 2022, PMID 36373573). The purposes of this study were to: (1) identify eCpG loci in one group oncology patients receiving CTX and (2) evaluate for differential methylation of distal regions of eCpG loci associated with AM CRF severity in an independent group of oncology patients receiving CTX.

## Study Sample

### Patients and settings

Oncology outpatients (n=1346) who were receiving CTX for breast, lung, gynecological, or gastrointestinal cancers were recruited as part of the previous grant (NCI R01 CA134900). Patients beginning their 2nd or 3rd cycle of CTX were assessed over two complete CTX treatment cycles (i.e., 6 assessments) using the Lee Fatigue Scale (LFS). In the week prior to their 2nd or 3rd cycle of CTX (i.e., Assessment), patients were asked to rate the severity of CRF in the past week upon awakening (morning CRF) and 30 minutes prior to bedtime (evening CRF). Blood samples were drawn for the omics analyses in EDTA and PAXgene collection tubes at enrollment. Differences in phenotypic characteristics between the groups were evaluated using the appropriate parametric and nonparametric tests.

### Instruments

Patients completed a demographic questionnaire, Karnofsky Performance Status (KPS) scale, Self-Administered Comorbidity Questionnaire (SCQ), and Alcohol Use Disorders Identification Test (AUDIT). Toxicity of the chemotherapy regimen were rated using the MAX2 index. Medical records were reviewed for disease and treatment information. The 18-item Lee Fatigue Scale (LFS) was used to assess physical fatigue and energy. Each item was rated on a 0 to 10 numeric rating scale. Mean scores were calculated for the 13 fatigue items. Higher scores indicate greater fatigue severity.

### Patient Samples

Of the 1346 patients who provided a blood sample, 115 were processed using microarray for gene expression (Illumina Infinium HumanHT-12 v4.0 Expression BeadChip) and DNA methylation (Illumina Infinium HumanMethylation450 BeadChip) and 584 were later processed using microarray for DNA methylation (Illumina Infinium MethylationEPIC BeadChip). *There was no overlap in patients between groups.*

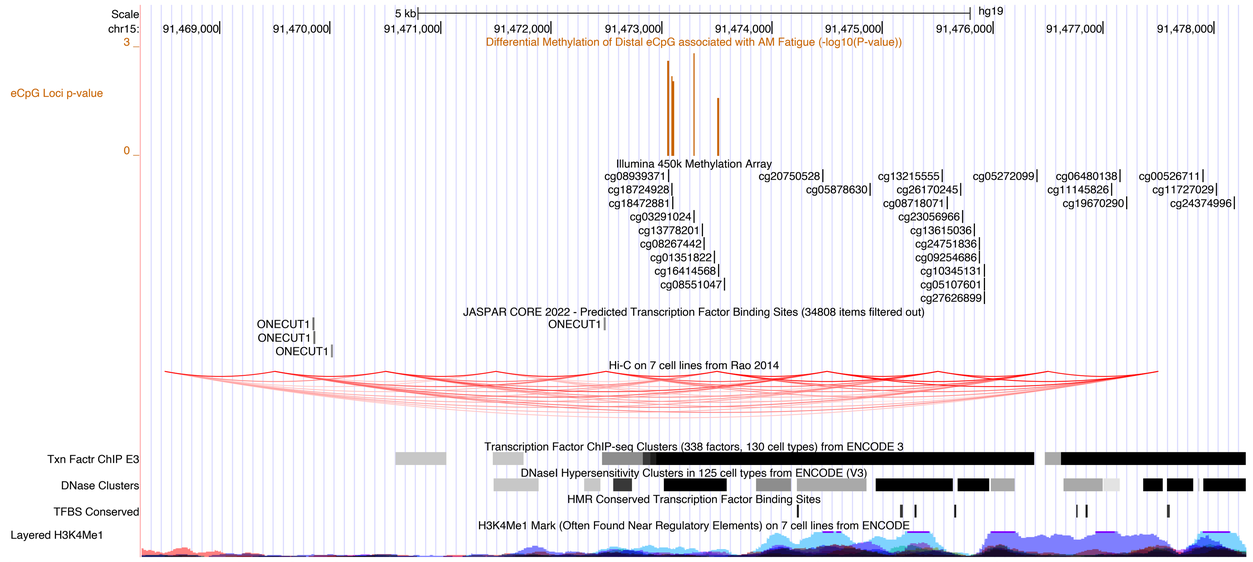


**Figure 1.** Screenshot of the University of California Santa Cruz Genome browser displaying the region on chromosome 11 of the hg19 (genome reference consortium Version 37) assembly of the human genome that includes the region #15_204 identified in this study as differentially methylated and the locations of the methylation probes included in the analysis. Assembly tracks show scale, chromosome, and position of the region. Predicted binding sites for the ONECUT1 transcription factor are shown in the JASPER CORE 2021 track. Putative regulatory regions are identified by the layered H3K4Me1, Transcription factor (ChIP-seq) clusters, DNase hypersensitivity clusters tracks. Hi-C track shows heatmaps of chromatin folding data (i.e., interactions detected between regions).

## Identification of Expression Associated CpGs (eCpGs)

The smaller group of patients (n=115) was used for the eQTM mapping analysis. A total of 30,952 gene expression probes and a total of 402,146 methylation probes were were included in the analysis. Gene expression were log-transformed and normalized. Methylation levels were quantified as M-values.

The Torch-eCpG tool (https://github.com/kordk/torch-ecpg, https://doi.org/10.1101/2023.03.07.531597, manuscript *In Review*) was used to map eQTMs. Age (years) was included in the model for the multiple linear regression. Distal eCpGs were identified with a CpG probe position >50K bp upstream of the transcript start site for the gene annotated for the expression probe. Putative eCpGs were identified having an uncorrected p-value < $1 \times 10^{-6}$. Mapping of all 12.4 $\times 10^{9}$ pairwise comparisons completed in ~1h using an NVIDIA A2 GPU (chuck sizes: n=100 expression probes, n=100,000 CpG loci).

**Table 1.** Multiple Logistic Regression Analysis of Demographic and Clinical Characteristics and Low and High Morning Fatigue Groups

| Oncology Patients (n = 585; Low MF 61.2% (n=361), High MF 38.8% (n=224)) | | | |
|---|---|---|---|
| Predictors | Odds Ratio | 95% CI | p-value |
| Age (years) | 0.97 | 0.95, 0.98 | $2.42 \times 10^{-5}$ |
| Karnofsky Performance Status score | 0.95 | 0.93, 0.97 | $1.36 \times 10^{-9}$ |
| Self-Administered Comorbidity Questionnaire score | 1.08 | 1.01, 1.15 | 0.030 |
| Married/partnered (yes) | 0.63 | 0.43, 0.93 | 0.021 |
| Exercise on a regular basis (yes) | 0.73 | 0.47, 1.11 | 0.136 |
| Cancer Diagnosis | | | |
| Breast | 1.00 | | |
| Gastrointestinal | 0.51 | 0.33, 0.81 | 0.004 |
| Gynecological | 0.77 | 0.48, 1.45 | 0.534 |
| Lung | 1.77 | 0.36, 1.38 | 0.322 |
| Overall model fit: AUC of the ROC = 0.751 | | | |

Abbreviations: AUC = area under curve; CI = confidence interval; CTX = chemotherapy; kg = kilogram; $m^2$ = square meter; MF = Morning Fatigue; ROC = receiver operating characteristic; RT = radiotherapy

**Table 2.** Summary of Differentially Methylated eCpGs Region between oncology patients receiving chemotherapy in the Low and High Morning Fatigue Groups.

| Region | CpG loci | logFC | P-value | Region Statistics | eCpG Mapped Genes |
|---|---|---|---|---|---|
| 15_204 | cg03291024 | 0.168 | 0.001 | FDR < 0.0002 | One Cut Homeobox 1 (ONECUT1) |
| chr15: | cg08939371 | 0.171 | 0.002 | | CAP-Gly Domain Containing Linker |
| 91473291- | cg16414568 | 0.139 | 0.027 | | Protein 1 (CLIP) |
| 91473091 | cg18472881 | 0.178 | 0.009 | | Solute Carrier Organic Anion Transporter |
| | cg18724928 | 0.174 | 0.006 | | Family Member 3A1 (SLCO3A1) |

## Evaluation of Differentially Methylated Regions

In the larger group of patients (n=584 total, n=224 High and n=360 Low AM CRF), methylation was measured at 794,441 loci. Of these, only those that were identified as an eCpG were evaluated for differential expression in this analysis. Methylation levels were quantified as the M-score. The final regression model included demographic (i.e., age, partnered, exercise) and clinical (i.e., KPS, cancer type, SCQ) characteristics that were significantly different between the two groups as well as three surrogate variables identified by SVA.Tests for differential methylation for eCpGs between groups were performed using the limma package for R. Differentially methylated distal regions were identified using a sliding window approach (window size=5000 bp, $\geq$2 distal eCpGs). Fisher's method was used to test for significant DMRs and assessed with a false discover rate (FDR) for significance (FDR < 0.05) and as suggestive (FDR < 0.10).

## Results

Of the 25,310 Distal eCpGs identified in the eQTM mapping analysis ($p < 1.0 \times 10^{-6}$), 9,656 were evaluated in the DM analysis and included in the DMR analysis.

The sliding window analysis identified 1285 distal regions. One region (#15_204) was found to be significantly differentially methylation (FDR < 0.0002)(Table 2). Three genes were associated with the CpG loci. Two (#5_151 and #6_194) were suggestive (FDR <0.094)

## Discussion

This study was the first to perform an eQTM mapping analysis in oncology patients receiving CTX. This study is also the first to identify differentially methylated regions associated with AM CRF severity in oncology patients receiving CTX.

Of the three genes mapped to CpGs in region 15_204, ONECUT1 is a transcription factor (TF) (Figure 1). This TF antagonizes glucocorticoid-stimulated gene transcription, glucose metabolism, and cell cycle regulation. We identified 437 target genes for this TF using the DoRothEA database (https://saezlab.github.io/dorothea/), which together represent a significantly enriched protein-protein network (STRING, $p < 1.0 \times 10^{-16}$). A second gene, SLCO3A1, is predicted to enable sodium-independent organic anion transmembrane transporter activity. Involved in positive regulation of NF-kappaB transcription factor activity; positive regulation of protein phosphorylation; and prostaglandin transport. In terms of CRF, genetic variations in SLCO3A1 are associated with the occurrence with myalgic encephalomyelitis/chronic fatigue syndrome.

While this study has a relatively large sample size and crosses two independent samples and two molecular datatype, our findings require independent validation. Given the sample was primarily white, female, and well educated, future studies should include a more heterogenous sample of patients. Longitudinal studies are needed to assess for associations between changes in CRF and changes in gene expression and methylation. Functional analyses of the effects of changes to the methylation of the region on gene expression are needed. Finally, the eQTM mappings will be made available for use by other groups.

This approach has identified eCpGs in oncology patients receiving CTX and identified a significantly differentially methylated region associated with AM CRF severity. These findings can be useful to identify novel therapeutic targets. These methods may be generalizable to other symptoms and patient-reported outcomes associated with cancer and its treatment (e.g., sleep disturbance, depression, anxiety, dyspnea, pain).

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